

Predicting Solar Flares

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The purpose of this project is to take in images taken from satellites orbiting the sun before a solar event and use them to predict whether or not a solar flare will occur.

Dataset Summary

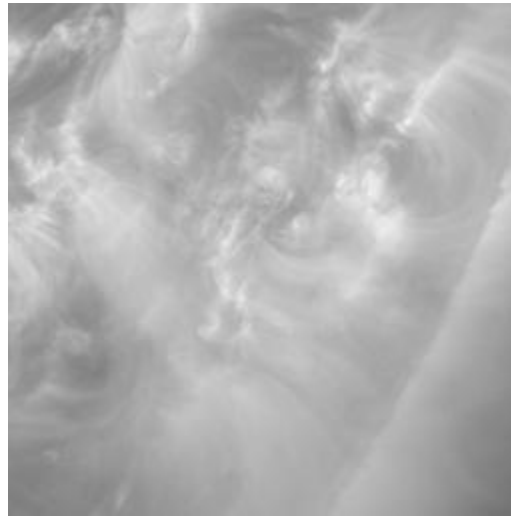


Figure 1: Example image from the dataset.

The dataset that will be used is the SDOBenchmark. This dataset consists of data four times before a 24 hour prediction period: 12h, 5h, 1.5h, and 10 minutes before. Each time sample has ten images- 8 UV wavelength images and 2 magnetic field images. All images are 256x256 jpegs.

This dataset is from Kaggle

(<https://www.kaggle.com/datasets/fhnw-i4ds/sdobenchmark>)

This dataset contains 8,336 training samples and 866 test samples.

This dataset contains 40 features (images of different types and sample periods)

The target variable is the peak emission of the sun within 24 hours (between $1e-9$ and $1e-3$).

This dataset is appropriate for our project because it is actually providing multi-wavelength solar imagery paired with labeled flare outcomes, making it well-suited for a supervised image-based prediction task.

Each team is required to submit an **Iteration 2 Proposal** describing the progress made since Iteration 1. At this stage, your dataset should be fully prepared, and your team should be ready to implement and evaluate your machine learning model(s).

Data Preprocessing

The preprocessing steps that will be taken are as follows:

1. Images will be resized to 128x128 (from 256x256) to save computation
2. Pixels will be normalized to a [0,1] scale
3. Missing images- missing images will be passed as all 0 with a flag describing them as missing.
4. A logarithmic scale will be applied to the output to better reflect the nature of the phenomena.
5. There are no duplicate records in the dataset, so no deduplication was required.
6. Outliers correspond to rare large-flare events, which are real and scientifically meaningful so they are retained here instead here rather than that being removed.
7. No categorical encoding is needed since all inputs are image pixel arrays rather than categorical attributes.
8. Feature selection is planned during training by removing redundant or low-importance image types to test whether it improves model performance.

These steps may be changed during training.

Machine Learning Model

A **Convolutional Neural Network (CNN)** will be used on this data. This is because this is an image classification problem, and CNNs are relatively simple to set up and we have used them in class.

The biggest advantage of a CNN is that it can automatically learn features from pixels without additional preprocessing. The hyperparameters that will be tuned are the **learning rate, number of blocks, batch size, dropout rate, weight decay, and loss function**.

Questions

1. Which image types are the most useful for predicting solar flares?

2. Which patterns tend to appear before a solar flare?
3. What features are best at predicting whether a solar flare will occur?
4. How does hyperparameter tuning influence the accuracy of predicting solar flares?
5. How does data preprocessing influence the accuracy of predicting solar flares?

Data Understanding

There are 8,336 training samples and 866 test samples. This means that there are roughly 128 x 128 pixels x 40 images, or around 655,000 “features”.

The dataset is very imbalanced, as quieter flares are much more common than larger flares. This is because the sun having large solar flares is rare.

There are many missing values in the dataset, as in some time steps only some of the 10 image types exist (the data was not taken or recorded accurately). There aren't duplicate records, but there are outliers. These outliers are real events and will still be used.

Based on limited research, solar flares are driven by the magnetic field of the sun, so the magnetic field images would be expected to have the most impact.

Data Visualization

These questions will require the model to answer. By looking at which features have the most impact, the pixels that are having the most effect will be determined. These high-importance pixels will be analyzed to determine which attributes are strongly correlated.

Different image types will be compared to see if they correlate to each other to see if they contain intrinsic relationships.

Samples of high-flare and low-flare events will be compared to see which variables show the greatest difference between classes.

A correlation heatmap will be generated between image types to see if any are redundant or insignificant.

Data Analysis

We would expect the magnetic field images to be strong predictors due to solar flares being created by the sun's magnetic field.

We expect some of the 10 image types are redundant or correlated, as they observe different (and sometimes overlapping) temperature ranges of the sun.

We will remove some less important features (like some image types) to see if it improves model behavior.

We would expect preprocessing to improve accuracy by training on different levels of preprocessing, especially logarithmic vs linear outputs and different methods of handling missing data.